REAR-IR1: rear docking IR board, 1 3.3 V, 2 IR_L (PC8, TIM3_CH3), 3 IR_R (PC9, TIM3_CH4), 4 GND.
HALL-BIN1 / HALL-TOWER1: Hall boards, SH 3-pin (ZX-SH1.0-3PLT, so a Hall cable cannot go into a 4-pin socket): 1 3.3 V, 2 OUT (PE4 / PE5, low = magnet near), 3 GND. All mounting pads to GND.

## CLIFF
File: CLIFF.kicad_sch

## MAIN BRUSH MOTOR
File: MAIN-BRUSH-MOTOR .kicad_sch

## WHEEL MOTORs
File: WHEEL-MOTORs .kicad_sch

## MAIN FAN
File: MAIN-FAN .kicad_sch

## SIDE BRUSH MOTORs
File: SIDE-BRUSH-MOTORs .kicad_sch

## BUMPER
Bumper Sensors
File: bumper_sensors.kicad_sch

## MCU-STM32
File: STM32G473.kicad_sch

## CARPET-SENSOR
File: Carpet-sensor.kicad_sch

## WD
WATCHDOG
File: WATCHDOG.kicad_sch

## LiDAR
File: LIDAR.kicad_sch

## FRONT SENSORS
File: Front-Sensors.kicad_sch

## IMU
File: IMU-ICM-4267-P .kicad_sch

## BUTTON-LEDs
File: BUTTON-LEDs .kicad_sch

## POWER CONTROL
File: SYSTEM-POWER.kicad_sch

## BATTERY CHARGER
File: Battery-Charger.kicad_sch

| | | |
|---|---|---|
| Sheet: / File: Main.kicad_sch | | |
| Title: | | |
| Size: A3 | Date: | Rev: |
| KiCad E.D.A. 10.0.4 | | Id: 1/16 |

5V USB, SENSORS, LIDAR POWER RAIL
– set 5V_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode: Fsw 418 kHz (RT 232k)

U22 TPS54560BDDAR
VBAT-14V4
PWR_FLAG
VSENS_5V_USB_END
C119 100nF/50V
10 R146
L3 CYA0630-3_3UH
VSENS-5V-USB
D43 SS56
D46 YLED0603B
C115 10uF/50V
C116 10uF/50V
C117 100nF/50V
R206 232k
R168 100k
R147 4.7k
C120 47nF/50V
R148 12k
R149 2.2k
C121 10uF/10V
C122 10uF/10V
C147 10uF/10V
1k R89

MOTOR POWER RAIL 14.4V

– on/off motor power, no soft ramp
– set VMOT-14V4-EN LOW to enable power
  – floating (MCU flash/reset) disables power
– WD_OK LOW shuts power off

VMOT-14V4: Q1 + Q37 AO4407C in parallel
– ~7.5 A normal (with the MSD-G fan) is ~0.25 W each
– short stall peaks (~13 A) fine with copper under the drain
– Vgs +/-25 V, no gate divider needed

VBAT-14V4
Q1 AO4407C
PWR_FLAG
VMOT-14V4
R165 10k
R36 10
C62 10uF/25V
D37 SS34
Q37 AO4407C
R222 10
VMOT-SW-G
VMCU-3V3
10k R184
Q19 AO3400
WD_OKD
VMOT-14V4-END

3.3V MCU POWER RAIL

– POWER_EN is 14.4V
– set POWER_EN HIGH to enable

TPS54560B (U21, U22):
– non-synchronous
– COMP 4.7k + 47nF (all-ceramic Cout, no pole cap)
– SS56 catch diode: fsw 418 kHz (RT 232k)
– EN abs max 8.4 V: R207/R208 on U21

U21 TPS54560BDDAR
VBAT-14V4
C111 100nF/50V
10 R141
L2 CYA0630-3_3UH
PWR_FLAG
VMCU-3V3
R207 100k
R208 22.1k
C106 10uF/50V
C95 10uF/50V
C102 100nF/50V
R205 232k
D42 SS56
R144 10k
R145 10k
C112 47nF/50V
R143 4.7k
C113 10uF/10V
C114 10uF/10V
C146 10uF/10V
D44 YLED0603B
1k R14

14.5V ULTRASONIC CARPET SENSOR POWER

L4 XRNR4030-10UH_M
D4 SS34_C8678
PWR_FLAG
VUSONIC-14V4
VSENS-5V-USB
U5 MT3608
NC SW IN GND EN FB
C149 10uF/10V
C148 10uF/10V
R160 232k
R159 10k
C150 10uF/25V
C151 10uF/25V
VUSONIC-14V4 = 0.6V x (1 + 232k/10k) -= 14.5V

STM-PWR-CTRL need to use for both EN and OFF from STM32 GPIO pin.
1- Hold EN to HIGH after buttone pressed
2- OFF board

VBAT-14V4
Q13 AO4407C
POWER-EN
R16 100k
R67 10k
C14 1uF/25V
R171 100k
PWR-SW-G
POWER-BTN
C22 1uF/25V
VMCU-3V3
R173 10k
PWR-BTN-SENSE
D26 1N4148
R172 10k
R190 1k
Q23 AO3400
STM-PWR-CTRL
PWR-LATCH
R193 1M
R191 100k
R192 47k
Q29 AO3400
C172 10uF/10V
C173 100nF/50V
U23 74LVC1G17
Q30 AO3400

Hardware off timer: holding the power button for ~4–9 s turns the MCU rail off even if the firmware hangs (Q29 releases C172, U23 trips, Q30 pulls PWR-LATCH low), Firmware turns off at 3 s.
While charging, the rail-hold (Q31 on BATTERY-CHARGER, from DC_IN) keeps the MCU rail on regardless, and charger power alone turns the MCU rail on (Q31 pulls PWR-SW-G low).

VMCU-3V3
Q16 AO3401
VSENS-3V3
PWR_FLAG
100k R99
C176 1uF/25V
D45 YLED0603B
1k R98
VSENS_3V3_END
10k R195
C64 100nF/50V

3.3V SENSORS POWER RAIL

– C176 keeps Q16 off when power switches ON
– C64 soft-ramps power
– set VSENS_3V3_EN LOW to enable sensors power

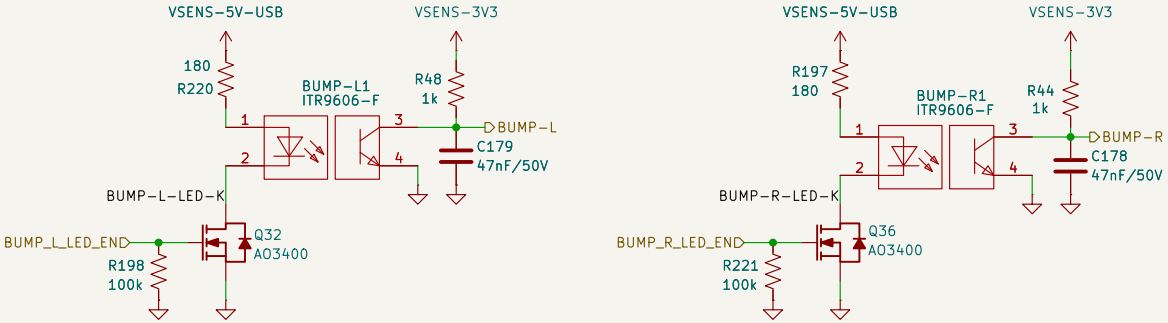
MCU VBAT
– Low quiescent current 2.5µA
– Zener diode shuts off when main battery is low

VBAT-14V4
D2 ZMM9V1_C2891750
U16 HT7533-1_C14289
Vout Vin GND
PWR_FLAG
VMCU-BAT-3V3
C140 10uF/25V
C141 10uF/25V

PCB mounting standoffs (Wurth WA-SMSI, M3), moved here from the removed M.2 sheet.

MP1 MP2 MP3 MP4 MP5 MP6

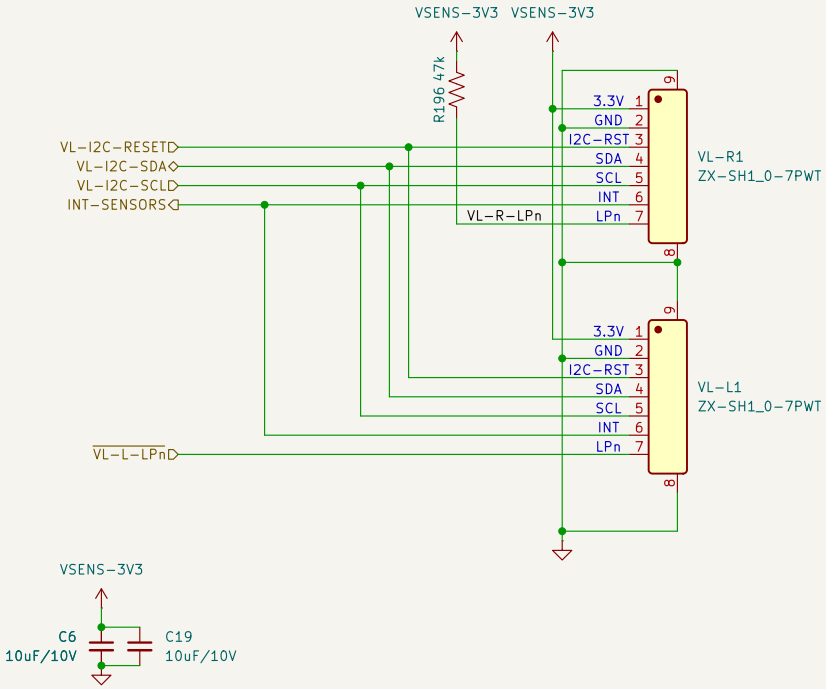
Sheet: /POWER/
File: SYSTEM-POWER.kicad_sch
Title:
Size: A2 | Date: | Rev:
KiCad E.D.A. 10.0.4 | Id: 2/16




- slotted photo-interrupters
- BUMP-X reads low when light passes the slot
- drive BUMP_L/R_LED_EN PWM <<15 usec period to enable
- tune R44, R48, LED PWM duty to let ADC sense full open/close range

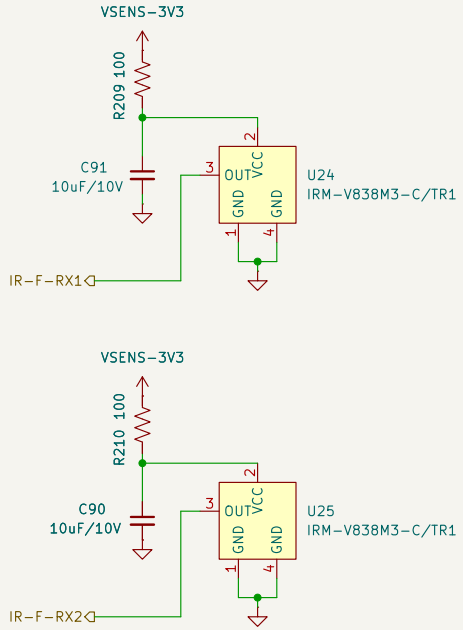
| | |
|---|---|
| Sheet: /Bumper Sensors/ | |
| File: bumper_sensors.kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 3/16 |

## 2x VL53L7 ToF BOARDS

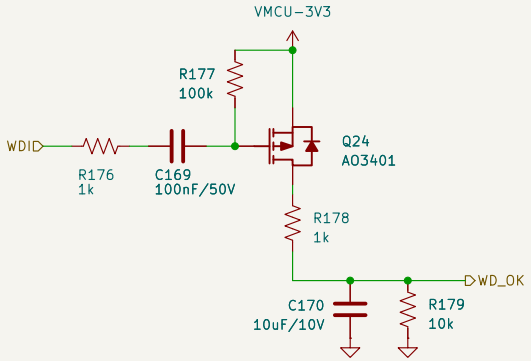



## 2 IR RECEIVERS

U24 left, U25 right (robot view). Side-looking: put both at the board front edge, lenses facing forward, ~10 mm apart, with a baffle between them in the bumper window. ~ 90 deg FoV




| | |
|---|---|
| Sheet: /FRONT SENSORS/ | |
| File: Front-Sensors.kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 4/16 |




RC CHARGE-PUMP WATCHDOG

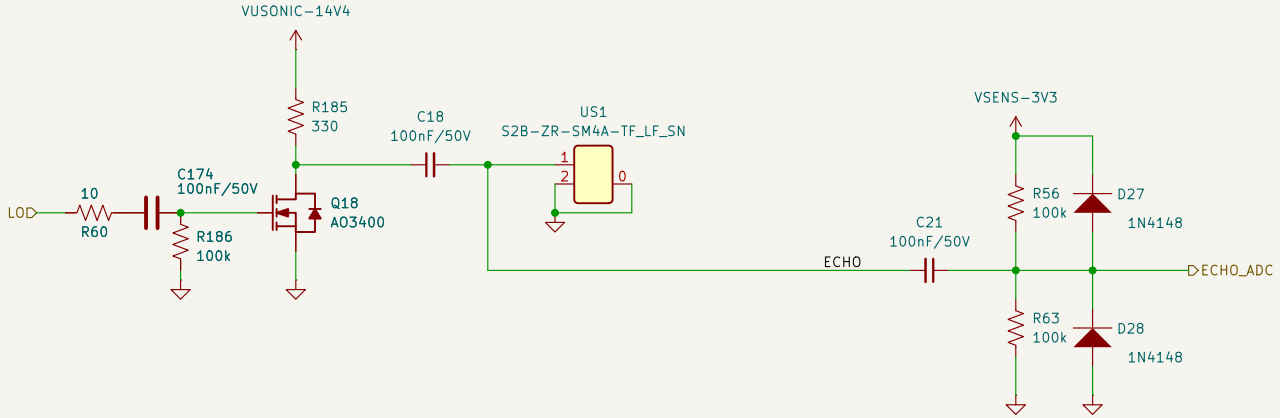
- WD_OK active HIGH
- WDI HIGH->LOW transition pets the dog

Firmware toggles WDI in software (50-1000 Hz, ~50% duty) only while the control loop is healthy.
Never drive WDI from a timer/PWM: a timer keeps running after the CPU hangs.
Each falling edge on WDI pulls the Q24 gate low through C169 and tops up C170 via R178.
WD_OK ~2.7 V while toggling. WDI stuck high, low or floating: <1 V in ~100 ms, <0.4 V in ~190 ms
(10 uF nominal; DC-bias derating shortens this). WD_OK starts at 0 V at power-up (fail-safe).
WD_OK drives N-FET gates only (14.4 V rail switch, LiDAR incl. laser): no resistive loads.
WDI pin: high-Z at reset, no pull-down.

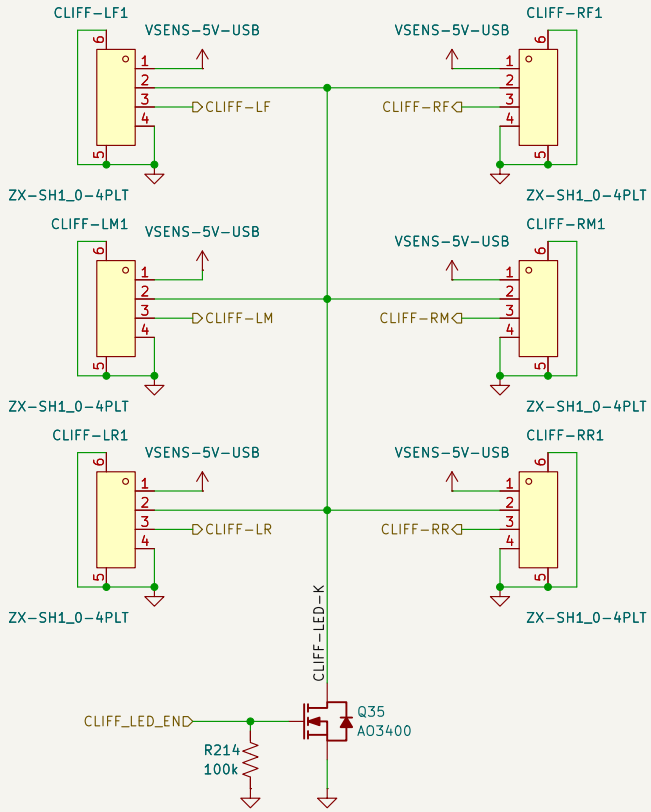
| | |
|---|---|
| Sheet: /WATCHDOG/ | |
| File: WATCHDOG.kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 5/16 |

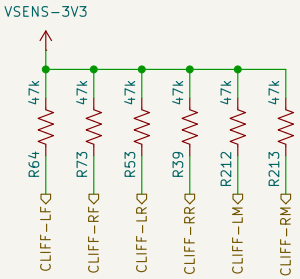
Carpet sensor driver: one low-side AO3400 + 330R pull-up from VUSONIC-14V4 (14.4 V). The P-FET half-bridge was removed: its 100k gate pull-up could not turn it off at 290 kHz (shoot-through). R186 holds LO low in reset/flash.
Drive LO in short bursts only: LO stuck high puts 0.63 W into R185.

C174 AC-couples LO to the Q18 gate: if LO sticks high, the gate decays through R186 (~10 ms) and Q18 turns off, so R185 is never stressed by firmware holding LO high.



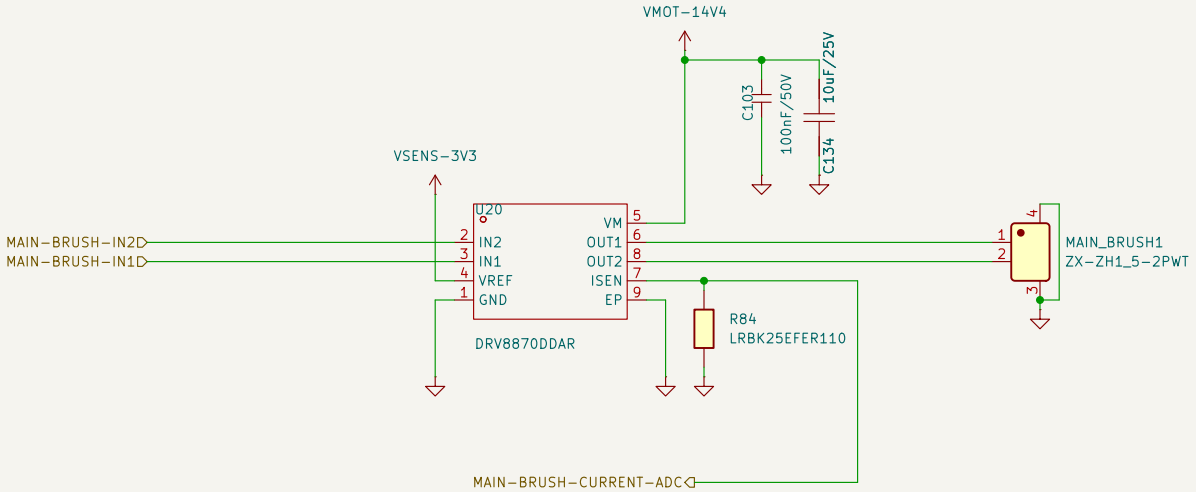

| | | |
|---|---|---|
| Sheet: /CARPET-SENSOR/ | | |
| File: Carpet-sensor.kicad_sch | | |
| **Title:** | | |
| Size: USLetter | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 6/16 |



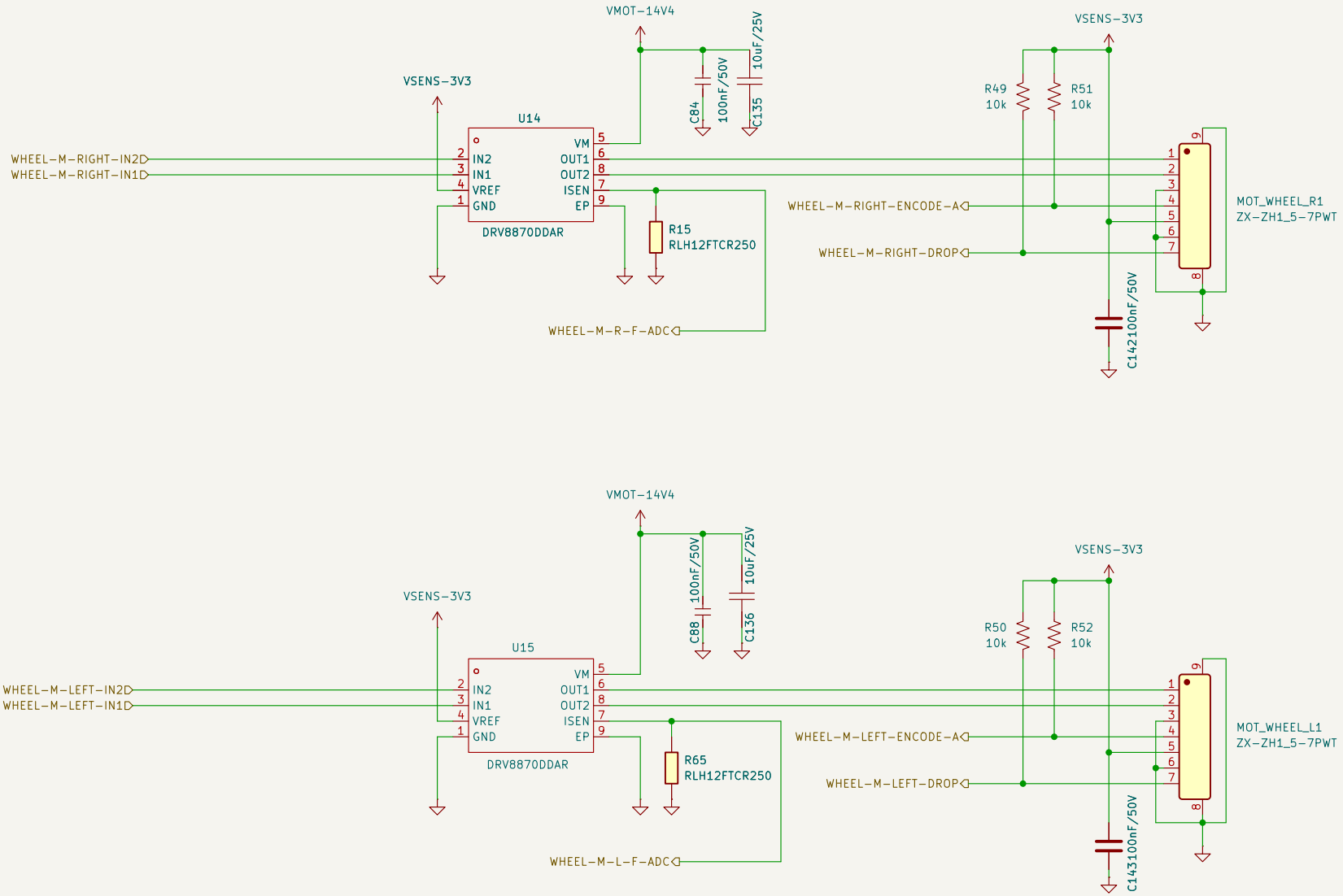




- kicad/cliff-sensor mating boards
- SH 4-pin vertical; use same (pin 1 to pin 1) cable
- set CLIFF_LED_EN LOW to disable
- modulate CLIFF_LED_EN, read each channel LED-on minus LED-off to cancel ambient light

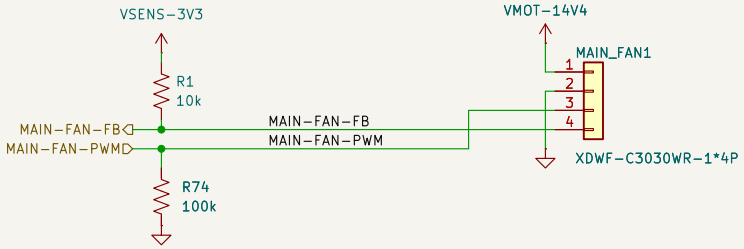
| | |
|---|---|
| Sheet: /CLIFF/ | |
| File: CLIFF.kicad_sch | |
| **Title:** | |
| Size: USLetter | Date: |
| KiCad E.D.A. 10.0.4 | **Rev:** Id: 19/16 |




| | | |
|---|---|---|
| Sheet: /MAIN-BRUSH MOTOR/ | | |
| File: MAIN-BRUSH-MOTOR .kicad_sch | | |
| **Title:** | | |
| Size: USLetter | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 22/16 |




| | |
|---|---|
| Sheet: /WHEEL MOTOR/ | |
| File: WHEEL-MOTORs .kicad_sch | |
| **Title:** | |
| Size: USLetter | Date: |
| KiCad E.D.A. 10.0.4 | Id: 23/16 |

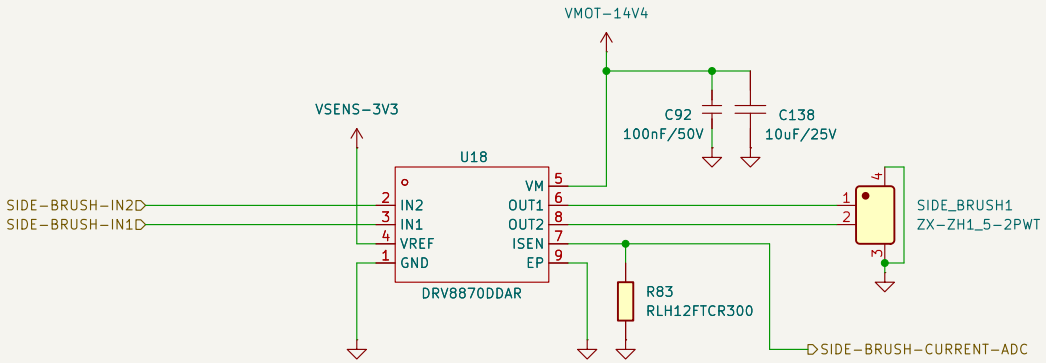



- FB feedback signal is open collector
- set PWM LOW to turn fan OFF
- MAIN_FAN1 matches Dreame L60 PE Ultra PCB

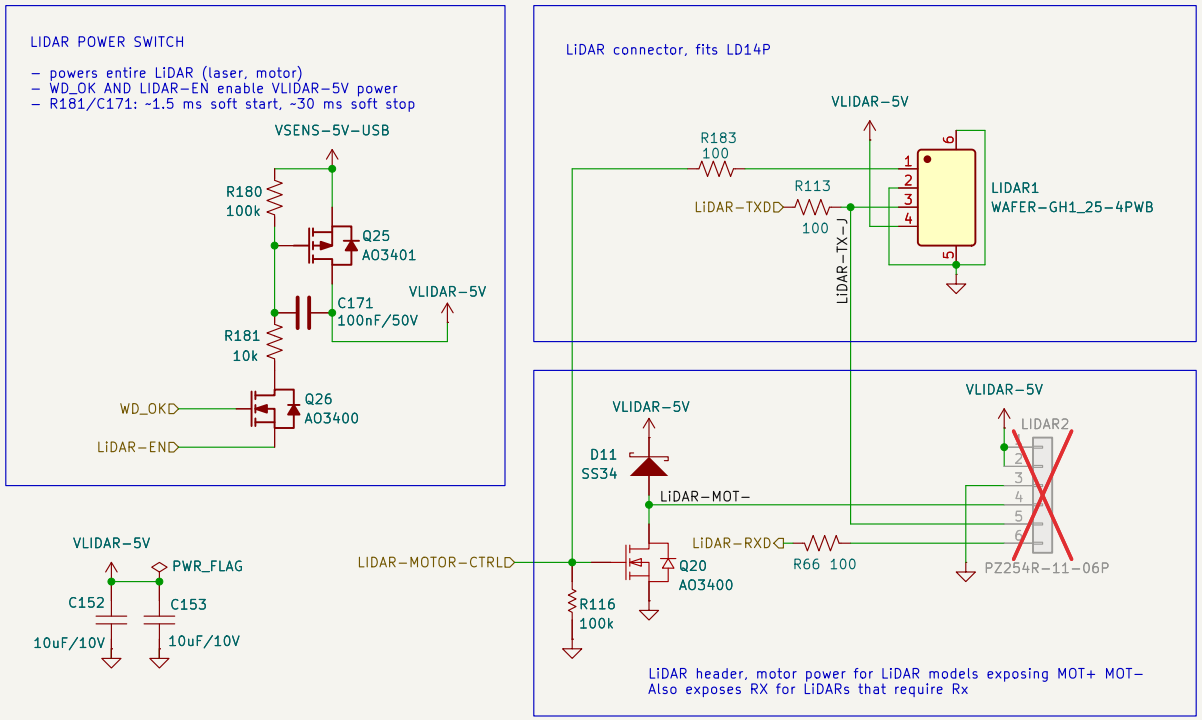



| | |
|---|---|
| Sheet: /MAIN-FAN/ | |
| File: MAIN-FAN .kicad_sch | |
| **Title:** | |
| Size: USLetter Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 24/16 |

# SIDE-BRUSH MOTOR CONTROL

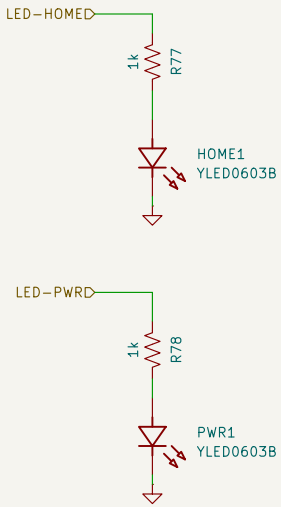


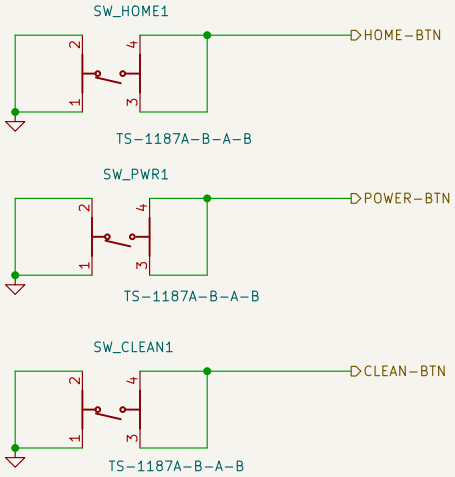

| | |
|---|---|
| Sheet: /SIDE-BRUSH/ | |
| File: SIDE-BRUSH-MOTORs .kicad_sch | |
| **Title:** | |
| Size: USLetter \| Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | Id: 25/16 |



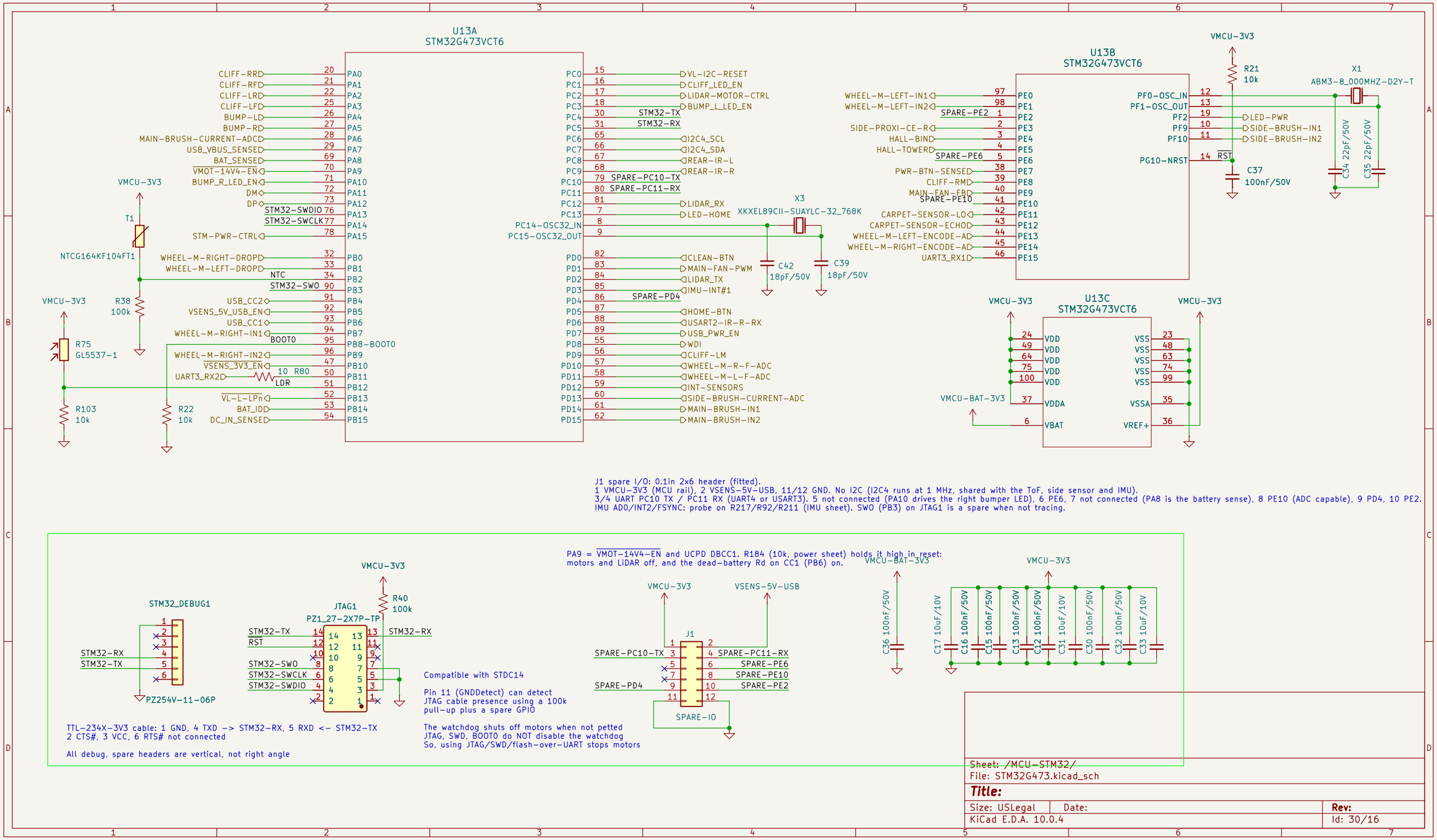

| | |
|---|---|
| Sheet: /LIDAR/ | |
| File: LIDAR.kicad_sch | |
| **Title:** | |
| Size: USLetter | Date: |
| KiCad E.D.A. 10.0.4 | Id: 26/16 |

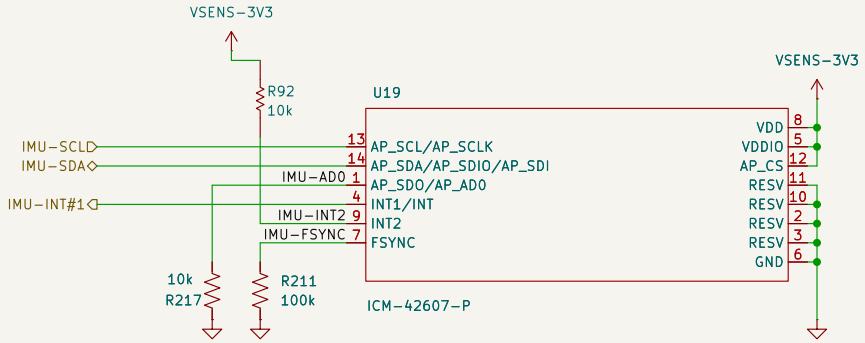
# LEDs AND BUTTONs



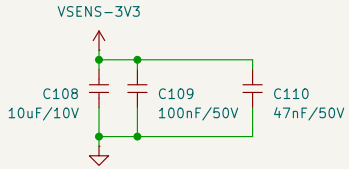




| | | |
|---|---|---|
| Sheet: /BUTTON-LED/<br>File: BUTTON-LEDs .kicad_sch | | |
| **Title:** | | |
| Size: USLetter | Date: | **Rev:** |
| KiCad E.D.A. 10.0.4 | | Id: 28/16 |


U13A
STM32G473VCT6
CLIFF-RR
CLIFF-RF
CLIFF-LR
CLIFF-LF
BUMP-LD
BUMP-RD
MAIN-BRUSH-CURRENT-ADC
USB_VBUS_SENSED
BAT_SENSED
VMOT-14V4-EN
BUMP_R_LED_EN
DM
DP
STM32-SWDIO
STM32-SWCLK
STM-PWR-CTRL
WHEEL-M-RIGHT-DROP
WHEEL-M-LEFT-DROP
NTC
STM32-SWO
USB_CC2
VSENS_5V_USB_EN
USB_CC1
WHEEL-M-RIGHT-IN1
BOOT0
WHEEL-M-RIGHT-IN2
VSENS_3V3_EN
UART3_RX2
VL-L-LPn
BAT_ID
DC_IN_SENSED
PA0
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13
PA14
PA15
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PB8-BOOT0
PB9
PB10
PB11
PB12
PB13
PB14
PB15
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC14-OSC32_IN
PC15-OSC32_OUT
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VL-I2C-RESET
CLIFF_LED_EN
LIDAR-MOTOR-CTRL
BUMP_L_LED_EN
STM32-TX
STM32-RX
I2C4_SCL
I2C4_SDA
REAR-IR-L
REAR-IR-R
SPARE-PC10-TX
SPARE-PC11-RX
LIDAR_RX
LED-HOME
CLEAN-BTN
MAIN-FAN-PWM
LIDAR_TX
IMU-INT#1
SPARE-PD4
HOME-BTN
USART2-IR-R-RX
USB_PWR_EN
WDI
CLIFF-LM
WHEEL-M-R-F-ADC
WHEEL-M-L-F-ADC
INT-SENSORS
SIDE-BRUSH-CURRENT-ADC
MAIN-BRUSH-IN1
MAIN-BRUSH-IN2
VMCU-3V3
T1
NTCG164KF104FT1
R38
100k
R75
GL5537-1
R103
10k
R22
10k
10 R80
LDR
X3
XKXEL89CII-SUAYLC-32_768K
C42
18pF/50V
C39
18pF/50V
U13B
STM32G473VCT6
WHEEL-M-LEFT-IN1
WHEEL-M-LEFT-IN2
SPARE-PE2
SIDE-PROXI-CE-R
HALL-BIND
HALL-TOWERD
SPARE-PE6
PWR-BTN-SENSED
CLIFF-RMD
MAIN-FAN-FBD
SPARE-PE10
CARPET-SENSOR-LO
CARPET-SENSOR-ECHO
WHEEL-M-LEFT-ENCODE-AD
WHEEL-M-RIGHT-ENCODE-AD
UART3_RX1
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
R21
10k
X1
ABM3-8_000MHZ-D2Y-T
C34 22pF/50V
C35 22pF/50V
PF0-OSC_IN
PF1-OSC_OUT
PF2
PF9
PF10
PG10-NRST
LED-PWR
SIDE-BRUSH-IN1
SIDE-BRUSH-IN2
RST
C37
100nF/50V
U13C
STM32G473VCT6
VDD
VDDA
VBAT
VSS
VSSA
VREF+
VMCU-BAT-3V3
J1 spare I/O: 0.1in 2x6 header (fitted).
1 VMCU-3V3 (MCU rail), 2 VSENS-5V-USB, 11/12 GND. No I2C (I2C4 runs at 1 MHz, shared with the ToF, side sensor and IMU).
3/4 UART PC10 TX / PC11 RX (UART4 or USART3). 5 not connected (PA10 drives the right bumper LED), 6 PE6, 7 not connected (PA8 is the battery sense), 8 PE10 (ADC capable), 9 PD4, 10 PE2.
IMU AD0/INT2/FSYNC: probe on R217/R92/R211 (IMU sheet). SWO (PB3) on JTAG1 is a spare when not tracing.
PA9 = VMOT-14V4-EN and UCPD DBCC1. R184 (10k, power sheet) holds it high in reset:
motors and LiDAR off, and the dead-battery Rd on CC1 (PB6) on.
STM32_DEBUG1
PZ254V-11-06P
JTAG1
PZ1_27-2X7P-TP
R40
100k
Compatible with STDC14
Pin 11 (GNDDetect) can detect
JTAG cable presence using a 100k
pull-up plus a spare GPIO
The watchdog shuts off motors when not petted
JTAG, SWD, BOOT0 do NOT disable the watchdog
So, using JTAG/SWD/flash-over-UART stops motors
J1
SPARE-IO
VSENS-5V-USB
C36 100nF/50V
C17 10uF/10V
C16 100nF/50V
C15 100nF/50V
C13 100nF/50V
C12 100nF/50V
C31 10uF/10V
C30 100nF/50V
C32 100nF/50V
C33 10uF/10V
TTL-234X-3V3 cable: 1 GND, 4 TXD -> STM32-RX, 5 RXD <- STM32-TX
2 CTS#, 3 VCC, 6 RTS# not connected
All debug, spare headers are vertical, not right angle
Sheet: /MCU-STM32/
File: STM32G473.kicad_sch
Title:
Size: USLegal
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 30/16

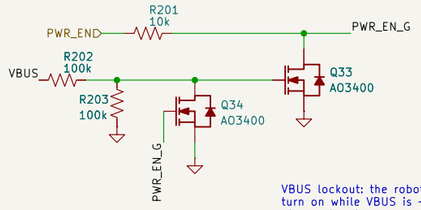



IMU on I2C4 (2.2k pull-ups R5/R6 on the top sheet). AP_AD0 pulled low by R217 (I2C address 0x68).
The IMU is leadless: probe AD0 on R217, INT2 on its pull-up R92 and FSYNC on R211. Run a wire to a spare pin if INT2 or FSYNC is needed.



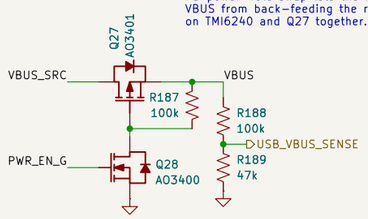

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| File: IMU-ICM-4267-P .kicad_sch | |
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| KiCad E.D.A. 10.0.4 | Id: 32/16 |

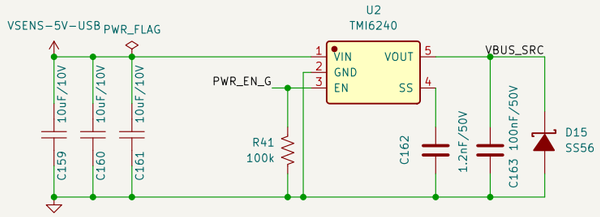
# USB-C

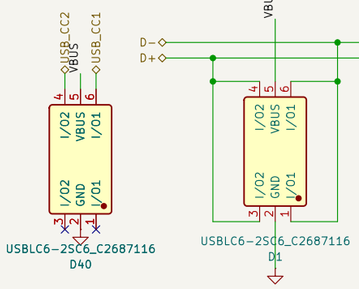


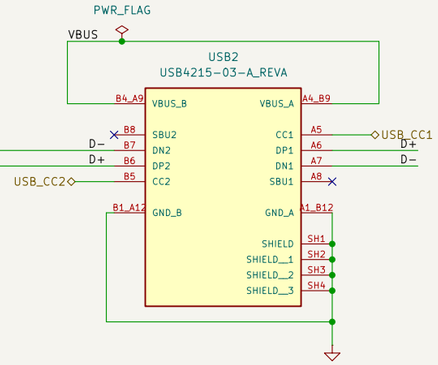

VBUS lockout: the robot 5 V (TMI6240 + Q27) can only turn on while VBUS is ~0 V. Q33 holds PWR_EN_G low if a phone already sources VBUS; once on, Q34 latches it.

USB-C (phone): phone is the USB host. CC1/CC2 go to the STM32 UCPD (PB6/PB4): Rd by default, PD power-role swap lets the robot charge the phone while it stays host. Q27 blocks the phone's VBUS from back-feeding the robot 5 V (the TMI6240 body diode conducts VOUT->VIN); PWR_EN turns on TMI6240 and Q27 together. TMI6240 now runs from VSENS-5V-USB (U25 buck removed).

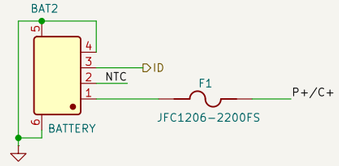


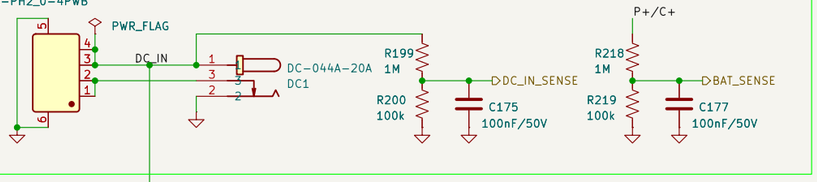

# BATTERY CONNECTOR



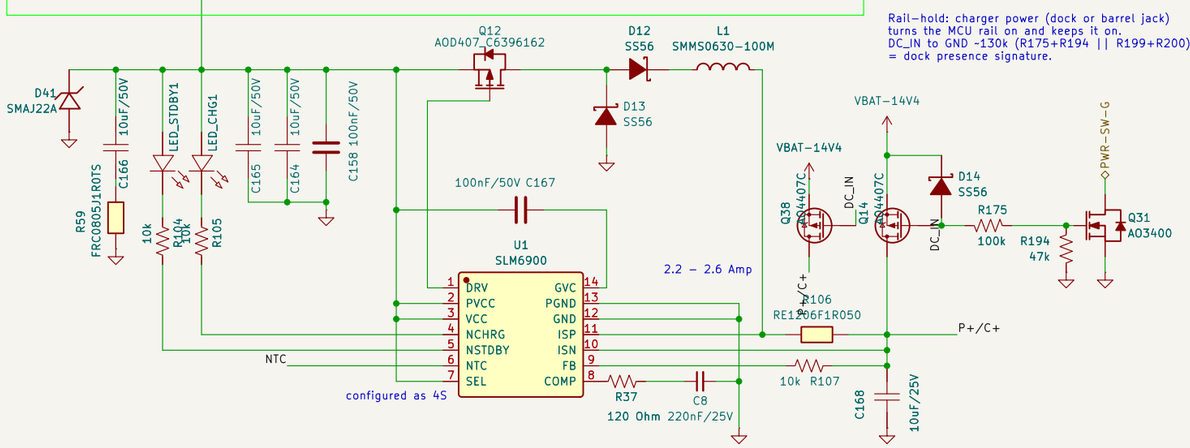

# BATTERY CHARGER

CONTACT_CHGR1
WAFER-PH2_0-4PWB

DC_IN: 20 V brick or dock. With the battery disconnected (Q14 off) VBAT-14V4 = DC_IN - 0.4 V, so the SLM6900 (95% max duty) needs >= ~18.5 V to finish a 4S charge. Keep DC_IN <= 22 V (D41, Q13 Vgs).




Rail-hold: charger power (dock or barrel jack) turns the MCU rail on and keeps it on. DC_IN to GND ~130k (R175+R194 || R199+R200) = dock presence signature.




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| Sheet: /BATTERY-CHARGER/ | |
| File: Battery-Charger.kicad_sch | |
| **Title:** | |
| Size: A3 | Date: |
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